

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Forest resources & deforestation - Set 1

Mark scheme

AQA spec 3.5.3; Biological resources; textbook pp. 360-369

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) • Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs • Food: fruit, nuts, bushmeat, livestock fodder; crops of forest origin (coffee, cacao, bananas, Brazil nuts); pigs and chickens were forest animals • Fibres: paper (wood pulp cellulose), cotton, viscose / rayon • Medicines: quinine (cinchona), taxol (yew), aspirin (willow); most species not yet researched	5
01.2	• Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time; total forest carbon (vegetation plus soil) is about double that in the atmosphere • Growing trees sequester CO₂ by photosynthesis and release oxygen; forest soils hold more carbon than the trees (especially boreal) • Planting / protecting forests removes CO₂ and reduces the enhanced greenhouse effect	3
01.3	• Highest transpiration of any ecosystem increases downwind rainfall; interception increases evaporation; deep organic soils retain water and delay runoff so river flow is more even and flooding reduced • Low albedo absorbs sunlight; heat stored and released at night reduces day-night temperature extremes; canopy microclimate: lower light, wind, higher humidity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	

Question	Marking guidance	Marks
02.1	• Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance; indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met • Single species: easy planting, thinning and felling over large areas; but each tree species supports its own community so few species overall • Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them • Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind; no old trees or dead wood for invertebrates and woodpeckers • Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species	6
02.2	• Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline • Continuing to grow them ties up land and capital for little extra timber; natural forests keep old and dead trees	2
02.3	• Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp • Wood fuel replaced by fossil fuels / electricity; so traditional management declined and woods became uniform	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
03.1	• South America: high loss ($5.1-5.5 \text{ Mha yr}^{-1}$) falling by about half by 2020 • Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest • Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking	3
03.2	• South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export; HEP reservoirs; roads improving access; mining • Africa: growing rural populations clearing land for subsistence farming; fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy); fodder cutting in dry areas • Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted; urban expansion	4
03.3	• They create barriers fragmenting wildlife populations • Improved access makes it easier to remove timber and farm produce, so clearance spreads along them	2
03.4	• Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades; slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
04.1	• Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned • Loss of canopy and litter exposes soil to rain splash and wind • Decomposing roots no longer bind soil so erosion, gullyng and landslides increase (especially on slopes) • Reduced organic matter and biota lower infiltration and water retention	4
04.2	• Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers • Increased albedo of cleared land (reflects more sunlight) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change	4
04.3	<i>Any 2 from:</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines • Loss of genetic resources / potential medicines before they are discovered	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
05.1	• Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow • Buffer zones with no harvesting or machinery near rivers / wetlands • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species	5
05.2	• The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management	3
05.3	• Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period • Disadvantage: more labour-intensive and expensive; more tracks needed per tree	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	
06.1	• B	1

Question	Marking guidance	Marks
06.2	<i>Any 4 from:</i> • Harvesting no faster than the MSY - resource never depleted • Mixed species plantations - more food choices, relationships and niches • Indigenous species - support indigenous insects, birds • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield • Selective logging - small clearings, less disturbance, long-term timber benefit • Retention of dead wood, buffer zones, corridors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	